

Branch Wars

Executive Command

Design & Technical Review

Revision 6 — findings, measured gameplay dynamics, and a record of what has now been fixed

What changed in this revision

Revision 1 read total market dominance as a runaway-leader defect. That was wrong — it is a deliberate setup for buying back into the market, and **Part 2 is rewritten around that intent**. New in this revision: the deposit-side economics that explain the whole balance problem, the commercial-banking model, an assessment of the strategy tree, and the UI work needed to make the comeback legible. **Revision 3** adds Part 9 on banking realism and product depth, and Part 10 on construction economics and information architecture.

Evidence base

Method	Scope
Full automated test suite	152 test groups executed individually
Headless engine experiments	~200 simulated campaigns across 4 scenarios and multiple seeds
Invariant sweep	1,753 turns checked for accounting and state violations
Formula analysis	Yield/loss rates derived directly from shipped engine constants
Live browser play-through	Full-stack campaign, all optional systems enabled
Source review	~14,900 lines across engine, UI, styles and content

Every quantitative claim was measured against the shipped `BRANCH_WARS.html` engine, not estimated.

Executive summary

The engineering is strong: 151 of 152 test groups pass, the balance sheet balances across 1,753 simulated turns, credit cohorts reconcile exactly. **The problems are design problems.**

One root cause explains most of the balance findings, and it sits in two lines of operations.js:

The root cause: deposits out-earn loans, risk-free

Deposits on Protect Margin net +4.17%/yr with no credit risk, no staff, and a **0% risk-asset weight** — and `depositIncome` is a pure function of the balance, so **you never have to lend them out to earn it**. A middle-market loan book nets +3.62%/yr *with* credit risk, a 100% risk weight, and staff and facility capacity consumed. Gathering deposits and refusing to lend is the mathematically correct strategy.

#	Finding	Severity	Effect
1	Deposits out-earn loans with zero risk	CRITICAL	The optimal bank is a vault.
2	Both policy levers have a dominant answer	CRITICAL	Aggressive deposits lose money in every regime.
3	The comeback arc is walled off	CRITICAL	Re-entry is blocked exactly where dominance happens.
4	Optional systems disable every ending	CRITICAL	12/12 games ran to the horizon with no result.
5	Business bankers produce no balance sheet	HIGH	Commercial banking is a fee annuity attached to nothing.
6	The loan book is a fixed point	HIGH	Converges to \$10M–\$25M regardless of effort.
7	The strategy tree is five spend meters	HIGH	One branch never used at all; average player reaches tier 2 of 4.
8	War room idle 72–90% of turns	HIGH	2 of 8 actions never fire.
9	The comeback is invisible in the UI	HIGH	11% share renders identically to 45%.
10	700px of chrome before any content	HIGH	No workspace visible unscrolled on a 720px screen.
11	LAN auto-detection has never worked	HIGH	Every two-computer game opens with a PowerShell error wall.
12	Commercial lending is disconnected from commercial banking	HIGH	Business bankers cannot originate a business loan.
13	No research or product development to speak of	HIGH	9 fixed SKUs, on/off switches, no upgrade path.
14	One loss type in the whole bank	HIGH	No fraud, disputes, rate risk or liquidity risk.
15	The flagship building is dominated by two basic ones	MEDIUM	Financial Center loses on cost, upkeep, staff and deposits.
16	Capacity preview reads +\$0K for every building	HIGH	The number you choose a building on is always zero.
17	Six fixed settings beat the hardest AI 85% of the time	CRITICAL	No situational judgment required to win.
18	The AI's own allocation guarantees morale decay	CRITICAL	Fixing it is the largest single lever in the game.
19	The war room moves the outcome by +1.4%	HIGH	The game's headline feature is statistical noise.

#	Finding	Severity	Effect
20	The campaign is 90% decided by month 48	HIGH	And in pilot mode it then never ends.
21	Four of eight mandates are unachievable	HIGH	Two require upgrade projects that are retired.
22	Cumulative earnings are negative; capex is 100% expensed	HIGH	Growth always reads as failure on the earnings line.
23	The variance is larger than the skill signal	CRITICAL	Identical players finish 658 apart; playing well is worth ~400.

How to read this document

Parts 1–8 cover the campaign design, economy and interface. **Part 9** covers banking depth and realism, **Part 10** construction and layout, **Part 11** measured gameplay dynamics — what actually decides a campaign — and **Part 12** the progression systems that were built and left wired in. **Part 13** records what has now been fixed, and **Part 14** is the prioritised roadmap.

Findings are numbered 1–46 and carry a severity chip. Every figure was measured against the shipped engine; simulation tables state their sample size. Where a claim rests on a single observation rather than a sweep, it says so.

If you read one section, read **Part 11** — and within it, **Finding 43**. It measures how much of a result is skill and how much is chance, and the answer reframes every other number in this document.

Part 1 · What is working

Listed first because several recommendations would damage these if applied carelessly.

Engine correctness is excellent

24 campaigns / 1,753 turns checking balance-sheet identity, credit-cohort reconciliation, stat ranges and save validity produced **zero violations**.

- **Vintage-based credit risk.** Each cohort carries the risk set at origination. Switching to Conservative does not retroactively repair loans you already made — correct, and unusual in a game.
- **Principal repayment is never counted as income.** Rigorous everywhere.
- **Capital tiers with escalating supervision.** WELL CAP → SUPERVISED → CONSENT ORDER → UNDERCAP → receivership. The best-designed system in the game.
- **Content variety is good.** 26 monthly events distributed 2.3%–5.2%, all five economic regimes appearing 15–26% across 960 simulated months.
- **Deterministic, dependency-free build** that refuses to compile if a source file is unwired.
- **The 'Review this month' panel** is the best screen in the game. Keep its pattern and reuse it.

The collapse pacing is right

Under original rules the decline is slow and dramatic — measured across 8 games:

Milestone	Median cycle
First market lost	13
Sixth market lost	47
Game ends	64

51 cycles between losing your first market and losing the game — over four in-game years. A re-entry branch costs \$560,000 and takes 3 cycles. There is enormous room for a comeback arc. The window already exists; Part 2 is about why it is closed.

Part 2 · The comeback arc

Design intent: total market dominance is a state, not an ending — the dominated player buys back into the market. That is realistic and it is the right call. The mechanic is implemented. It cannot be reached.

CRITICAL

1. Both halves exist, in different modes, and never meet

The re-entry mechanic is real and well-designed — `project-coordinator.js:160–184` grants a 15% minimum launch share, resets the exit streak, and gives a 4-cycle protection window (8 under regional operations, with share earned rather than granted). But:

	Original rules	Pilot mode (any optional system)
Total dominance occurs	Yes — 5/6 games, all 12 markets	Never — exited reset monthly
Market exit persists	Permanent, never cleared	Wiped monthly; barely registers
Re-entry grace code runs	No — gated behind p.accounting, which is false here	Yes
Rival franchise dividends	Active, forever	Disabled

Where dominance exists, re-entry is impossible. Where re-entry works, there is nothing to come back from.

Blocker A — the validator forbids it in as many words

`institution.js:201`, inside `projectTargetIssue` — the check that gates *project targeting*, not merely focus:

```
if(territory.exited && territory.exited[index])
return 'Your institution permanently exited that market. Choose an operating market.';
```

I forced a market exit and tried both paths in both modes. Focusing the market: blocked. Building a branch there: blocked by the same check. Nothing in the codebase ever sets `exited[i] = false` except the pilot's monthly wipe, `absorbFranchise` at game end, and `save migration`. The word “permanently” is literally accurate.

Blocker B — franchise dividends make dominance self-funding

Per exited market, per cycle, forever: $\text{value} \times \$6,000 \text{ cash} + \text{value} \times \$2,500 \text{ deposits}$.

Markets held	Free cash per cycle	Per year
3	\$540,000	\$6.5M
6	\$972,000	\$11.7M
12	\$1,578,000	\$18.9M

At full dominance that is **79× the loser's operating profit**, enough to fund 2.8 re-entry branches per cycle for the player who does not need them, and more than the entire origination capacity of a normal bank. This is the real snowball engine.

What to change

- **Replace the block with a price.** In `projectTargetIssue`, allow `kind: 'branch'` projects into exited markets at a premium scaling with the rival's share. Buying into a market someone owns outright *should* cost more — that is realistic challenger economics, and it turns “permanently exited” into “expensive to re-enter.”
- **Clear the flag on completion.** The grace code already resets `exitStreak` and sets `reentryUntil`; add `territory.exited[index] = false` so the market stops feeding the domination check and the rival's dividends.
- **Ungate the grace path** from `else if (p.accounting)` so it runs under original rules too.
- **Decay the franchise dividend** — taper it over ~12 cycles or cap it against the holder's own income. Dominance should be a strong position, not a money printer that outproduces the banking model.

With those, TOTAL MARKET DOMINATION stops being an ending and becomes what you designed: a near-death state with a long, expensive, genuinely dramatic road back. **It also makes the fixed-horizon scored finish load-bearing** — a fully recoverable game otherwise has no terminator at all.

CRITICAL 2. Optional systems disable every win condition

Every optional system requires `campaignRulesVersion: 1`, which enables pilot mode. Pilot runs this every month (`accounting-adapter.js:112`):

```
g.buyoutPressure=[0,0]; g.consolidationStalemate=0; // buyout can never accumulate
t.exited=[false,false]; // domination can never trigger
```

Ruleset	Games	Reached an ending	Median length	How
Original rules	8	8 of 8	67 months	Hostile buyout
Any optional system	12	0 of 12	never	—

Disabling buyout also silently killed two other systems: **Shareholder Defense** (0.0% usage at every difficulty — it requires buyout pressure) and, in practice, the entire **Acquisition Strategy** branch of the tech tree. Add a scored finish at a fixed horizon rather than re-enabling buyout, which would reintroduce the snowball.

Part 3 · The economic model

CRITICAL 3. Deposits out-earn loans, risk-free — the vault problem

Pilot mode forces `fundingRulesVersion: 2`, which excludes deposits from `riskAssets` entirely. Combined with the deposit spread formula:

	Deposits (Protect Margin)	Loans (middle-market, balanced)
Net margin	+4.17%/yr	+3.62%/yr
Credit risk	None	Yes
Risk-asset weight	0%	100%
Staff / facility capacity	None consumed	Both consumed
Requires lending to earn	No — pure function of the balance	—

This is the mechanical root of every balance finding in revision 1. Measured across 48 campaigns, a Conservative strategy (lend less, margin deposits, liquid capital) scored **1887** and never failed; max-lending scored **587** and failed **6 of 12** times. The simulation is correctly rewarding the strategy the formulas describe.

Fix. Deposits should be raw material, not a finished product. Either reduce `depositIncome` to a holding yield well below the loan spread, or make it depend on deployment — idle deposits earn a low securities rate, deployed deposits earn the loan spread. That single change turns “be a vault” into “gather deposits, then put them to work.”

CRITICAL 4. Both policy levers have one dominant answer

Deposit pricing — net spread per \$1 of deposits:

Regime	Protect Margin	Balanced	Win Deposits
Expansion	+4.27%/yr	+2.25%	−0.77%
Steady	+4.17%/yr	+2.28%	−0.55%
Tight money	+4.67%/yr	+2.12%	−1.67%
Downturn	+4.01%/yr	+2.33%	−0.21%
Recovery	+4.11%/yr	+2.30%	−0.44%

“**Win Deposits**” has a **negative net spread in every economic regime**. It multiplies income by 0.45 and cost by 1.2 — worse on both sides of the trade. Its only upside is a 1.25× growth multiplier on a balance that loses money per dollar held.

Lending standards show the identical shape — Conservative wins in every regime and every product (steady/consumer: 5.18% vs 3.47% vs 1.75%; downturn/consumer growth goes to **−2.25%**). `creditPolicy` scales losses while `loanMult` only scales volume — and volume is already capped.

Fix. Make the aggressive settings raise a *ceiling*, not just a rate. Win Deposits should raise deposit capacity and contest power; Growth lending should raise `loanCapacity`. Then both levers trade volume against margin instead of being strictly worse.

HIGH

5. The loan book is a fixed point, not a growth curve

Origination is a flow; the book is a stock that drains three ways. It stops growing where they balance:

$$\text{steady-state book} = \text{monthly originations} \div (1/\text{term} + \text{monthly loss rate})$$

Product	Term	Amortization	Losses	Drain	Book @ \$0.6M/mo	@ \$1.5M/mo
Mortgage	120 mo	0.83%/mo	0.19%	1.02%	\$58.9M	\$147.3M
Middle-market	48 mo	2.08%/mo	0.26%	2.34%	\$25.6M	\$64.1M
Consumer	24 mo	4.17%/mo	0.32%	4.48%	\$13.4M	\$33.4M

The hidden strategic fact

Product choice changes the ceiling by 4.4× — mortgages sustain \$58.9M where consumer sustains \$13.4M at the same origination rate and nearly identical net margin. But the catalogue sells Consumer as “*High-volume, high-yield production*”. It is high-volume *per month* and produces the *smallest* balance sheet. **The label teaches the wrong model.**

Fix. The engine already computes nextPrincipal and knows origination capacity. Show the ceiling: “*At your current capacity your book converges to about \$36M. You are at \$19M.*” That converts an invisible cap into a goal, and it is exactly how bankers think about a balance sheet.

MEDIUM

6. Operations bankers are the only staff who earn nothing

revenue = (a.service + a.business + a.lending) * 8000 + ... — a.operations is excluded, while expense charges \$18,000/month for every employee. An operations banker therefore costs \$26,000/month against the alternative and returns only a charge-off reduction worth roughly **\$2,250/month** on a typical \$12M book.

For operations staffing to break even on that channel alone the book would need to reach about **\$139M** — five times the highest ceiling in the table above. They pay for themselves only through compliance/score effects and the 2-person morale floor, which is a very indirect argument for a whole department.

Part 4 · Commercial banking

HIGH

7. Business bankers produce no balance sheet at all

Traced through operations.js:

```
depGain = (service*125000 + branch*85000) * depMult <- retail staff + branches only loanGain =
a.lending * 185000 * loanMult <- lending staff only commercialIncome = business*760 + merchant*650
<- business staff -> fees only
```

Business and merchant are relationship **counts**. They never touch deposits or loans. They also have **no attrition** — the only thing that removes one is a Commercial Client Raid, which fires in 2–4% of turns. They accumulate forever.

Measured over 6 campaigns / 72 months, all-in on each department:

Allocation	Score	Deposits	Loans	Biz rel.	Merch rel.	Monthly profit
All-in service	1662	\$61.0M	\$5.6M	166	115	\$69,543
All-in business	1637	\$55.5M	\$12.1M	221	202	\$235,493
All-in lending	1214	\$37.3M	\$17.2M	183	129	\$187,853
AI default	1612	\$54.0M	\$11.8M	204	178	\$145,296

At 221 business + 202 merchant relationships that is roughly **\$299,000/month of gross fee income** — against about **\$36,000/month** of net income from the *entire* \$12.1M loan book. Commercial banking currently out-earns the whole balance sheet by ~8× while consuming no funding, no capital, and carrying no credit risk. It is a free annuity attached to nothing.

Why separating the two books is the right fix

	Retail	Commercial
Deposits	Granular, insured, statistically sticky, rate-sensitive at the margin	Operating/DDA balances — the cheapest funding a bank has, sticky while the relationship lasts, but concentrated : one departure moves real money
Loans	Long amortizing (24–120mo), small, portfolio behaviour, frequent small losses	Shorter, floating-rate, lumpy ; revolving lines that draw and repay
Risk shape	Predictable, poolable	Concentration risk — a handful of names
Won and lost by	Price, convenience, advertising	Relationship, service, the banker

Adding businessDeposits / businessLoans driven by bizGain solves three problems at once:

- **It gives commercial deposits a job retail cannot do.** Operating balances are non-rate-sensitive, so a strong commercial franchise becomes your defence against a Targeted Deposit Campaign and against rate-shock funding cost — which finally gives the idle war room something to protect.
- **It breaks the lending ceiling honestly.** Commercial lines are relationship-driven, so a banker landing a \$2M revolver is not consuming retail origination capacity. This is the natural way past the \$10M–\$25M fixed point without inflating a retail multiplier.

- **It gives business bankers something to lose.** Add monthly relationship attrition so they must *retain*, not merely accumulate — which is also what makes Commercial Client Raid worth defending against, and what makes the Treasury specialization meaningful.

Part 5 · The strategy tree

HIGH

8. It is five spend meters, not a tree

Five branches × four tiers = 20 nodes, \$14.41M to complete. But `strategyLevel` is derived from *cumulative spend* crossing thresholds — there are no prerequisites across branches, no mutually exclusive routes within a branch, and no synergies. Structurally it is five parallel progress bars plus one binary specialization fork each (32 combinations).

Measured across 24 players over 90 months — total capability spend by branch:

Branch	Total spend	Avg per player	Specializations chosen
Branch Network	\$25,461,667	\$1,061,000 (tier 2 of 4)	Retail Density × 19
Digital Platform	\$11,337,313	\$472,000 (tier 1–2)	Customer Experience × 9, Automation × 7
Commercial Bank	\$5,863,495	\$244,000 (tier 0–1)	Specialized Credit × 4, Treasury × 1
Operational Excellence	\$3,480,000	\$145,000 (tier 0–1)	none
Acquisition Strategy	\$0	never invested in	none

- **The average player reaches tier 2 of 4 in their best branch** and tier 0–1 everywhere else. Capstones and roughly half the authored content are never seen in a normal campaign.
- **Acquisition Strategy — 4 nodes, \$2.23M of content — receives zero investment across 2,160 player-months.** It is 20% of the tree and it has never been played.
- **Retail Density is chosen by 79% of players**, so even the specialization layer has a default.
- The score treats it as a spend meter too: `strategyTotal(p)*18 + (capstone ? 30 : 0)`.

Turning it into an actual tech tree — my recommendation

Yes, this is worth doing, and it is the most natural place in the game to add depth. But the current problem is not that there are too few nodes — it is that **nodes are bought rather than chosen**. Adding 40 more spend thresholds would make it longer, not deeper. Four changes turn it into a tree:

- **1. Make tiers exclusive, not cumulative.** At each tier, offer 2–3 alternatives and let the player pick one — e.g. at Network tier 2, *Hub-and-Spoke* (fewer, larger offices, higher capacity each) versus *Dense Footprint* (many small offices, more market presence). Same cost, different bank. The specialization mechanic already proves this works; extend it to every tier.
- **2. Add cross-branch prerequisites.** A Corporate Bank capstone should require Digital tier 2 (treasury needs a platform); Serial Acquirer should require Operations tier 2 (you cannot integrate what you cannot run). This is what makes a tree a tree rather than five ladders, and it creates the opportunity cost that makes choices matter.

- **3. Fix the affordability curve first.** Deepening a tree the average player never gets halfway through makes the problem worse. Either lower tier costs, or — better — fix the economic model in Part 3 so a well-run bank generates enough profit to reach the interesting nodes.
- **4. Give Acquisition something to act on.** Its dead state is downstream of the disabled buyout machinery. Repoint it at the Part 2 re-entry system — cheaper re-entry, faster establishment, better launch share — and it becomes the comeback branch, which is thematically perfect.

Sequencing

Do the tech tree **after** Part 3 and Part 4. The tree multiplies whatever economy it sits on; deepening it now would mostly add more ways to buy score. Once deposits, lending and commercial banking each have real trade-offs, the tree has something to differentiate.

Part 6 · Competition and the map

HIGH

9. The war room is idle most of the game

AI action selection across 2,880 generated plans:

Difficulty	Hold Position	Retention Def.	Commercial Raid	Relationship Def.	Talent Raid	Deposit Raid	Shareholder Def.
Analyst	89.9%	7.2%	1.5%	1.3%	0.1%	0.1%	0.0%
Strategic	71.8%	23.6%	2.2%	2.2%	0.2%	0.0%	0.0%
Chairman	77.2%	16.6%	4.0%	2.0%	0.2%	0.1%	0.0%

- **Attacks fire in under 5% of turns.** The headline feature of the game's name is dormant.
- **Shareholder Defense never fires at all** — it requires buyout pressure, which pilot zeroes.
- **Difficulty is not monotonic:** Strategic acts more often than Chairman, because the affordability gate (cash – cost ≥ 220000) binds differently.
- **Cause:** attacks cost \$180K–\$260K plus a \$220K reserve — roughly \$400K–\$480K of free cash against ~\$20K/month of early profit. Scale costs to bank size instead of fixed dollars.

MEDIUM

10. The city map barely affects deposits

depositContest sums competitive pressure across *all* markets into a single figure, picks one winner, and transfers one lump sum. Per-market detail is used only to name the largest contributor in the log. **You can win five markets, lose one badly, and end up net negative.** Local positioning does not produce local outcomes.

Compounding it, competitionScale reduces movement on larger maps: **12 markets** → **×0.6**, 8 → ×0.86, 6 → ×1.0. A bigger map produces *less* interaction, which is backwards for a game about territory.

MEDIUM

11. Half the monthly events reduce to one axis

26 events with better variety than first appears — about 14 have genuinely distinct axes. But 12 of 26 are “spend cash” versus “don’t”, and there is a structural asymmetry:

Option	What the player is shown
Paid branch — “Emergency vendor support”	Exact figures: <i>Immediate cash paid \$160,000 · Cash change −\$160,000 · Equity change −\$160,000</i>
Free branch — “Recover internally”	No numbers at all. <i>“Technology and Operations determine the damage.”</i> Reported afterwards as an unquantified sentence: <i>“suffered extended service disruption.”</i>

The free option is unquantified, so the decision cannot be evaluated. Six events use the “X determines Y” pattern. Give the free branch an expected range.

Part 7 · Errors and correctness

The test suite is healthy: **151 of 152 groups pass**. The single failure points at a real user-facing bug.

HIGH

12. LAN auto-detection has never worked (verified fix)

OPEN_BRANCH_WARS.bat escapes pipes as ^| inside double-quoted PowerShell arguments. Inside quotes cmd does not treat ^ as an escape, so PowerShell receives literal carets and fails to parse. Captured from a stubbed run:

```
ARGS: -NoProfile -ExecutionPolicy Bypass -Command "Get-NetRoute -DestinationPrefix '0.0.0.0/0' ^|
Sort-Object { ... } ^| ..." PowerShell: Unexpected token '^' in expression or statement. Result:
"Could not detect a LAN address. The game will open normally."
```

Verified fix — after removing the carets, detection succeeded immediately (192.168.0.87):

```
sed -i 's/\^|/|/g' OPEN_BRANCH_WARS.bat
```

Also fix the test: tests/portable-launcher.test.js stubs powershell.cmd in the working directory, which only resolves when NoDefaultCurrentDirectoryInExePath is unset. Put the stub on PATH — that is why this survived every prior release gate.

MEDIUM

13. Smaller defects

- **The campaign-size selector is a dead control.** Any optional system forces scope: 'regional' (6 markets) at campaign.js:42, but the “National · 12 markets” dropdown stays enabled. A status line discloses it, so it is not silent — but disable the control.
- **“Market Leader” is awarded to both players at once.** The key is firstMarket (“Control your first market”) but the display name reads as an exclusive rank. Rename it **First Market**. The message also omits the +3 influence it actually grants.
- **Reported loan growth contradicts the book.** Over 12 instrumented months loanGrowth read −\$31,858 falling to −\$118,422 while loans grew \$9.4M → \$15.5M. Rename it or report gross origination separately.
- **Plural agreement:** “1 relationships left”, “1 edited instructions”, “1 quarter-work units” — nine hardcoded plurals; seven other files already do this correctly.

- **Raw floats in player copy:** “0.978 quarter-work units uncovered · Credit administration: 4.3 · Technology: 1 · Risk: 1.195” — three decimals, inconsistent precision in one sentence.
- **All 24 optional systems are labelled “preview”** — hardcoded maturity: 'preview'. A label that never varies carries no information.

Part 8 · The interface

HIGH

14. The comeback arc is invisible on the map

I searched all of `src/ui/` for the exit mechanic. Outside one line, it does not exist in the interface.

What exists: `session.js` builds a proper countdown — “*MARKET EXIT WARNING · Downtown 2/3 · Industrial 1/3 below the 12% survival line.*” But its severity is 'watch', the same tier as a routine intel note, while capital distress and funding exposure get 'hot'. And it lives in the **Intelligence** workspace — behind the chrome wall and two navigation clicks.

What does not exist at all: `grep` for `exitStreak`, `exited`, `re-entry` across every UI module returns nothing. The map has no idea any of this is happening — and its whole vocabulary is three states:

```
ownerLabel(share) -> share>=55 ? 'YOU CONTROL' : share<=45 ? 'RIVAL CONTROLS' : 'CONTESTED'
```

A market at 11% share — one cycle from being lost forever — renders identically to one at 45%. Same colour, same tile. The most dramatic moment in the design has no visual representation.

State	Engine field	UI today	Should be
Slipping	<code>exitStreak 1/3</code>	nothing on map	Amber edge on the tile, “1 of 3”
Critical	<code>exitStreak 2/3</code>	nothing on map	Red pulse, 'hot' severity, blocker in Review-this-month
Withdrawn	<code>exited[i]</code>	nothing	Greyed tile with the re-entry price on it
Re-entering	<code>reentryUntil</code>	nothing	“Establishing — 4 cycles” badge, protected marker

Put the re-entry cost on the tile. A greyed district reading “*Re-enter: \$560,000 · 3 cycles · 15% launch share*” turns a dead square into the thing you are saving up for — which is the entire point of the design. Show the rival's franchise dividend there too (“*Downtown pays your rival \$90,000/cycle*”) so the loss is legible and the comeback has a stated reason.

HIGH

15. 700px of chrome before any content

Measured on a 1280×720 viewport with the plan review collapsed: the first workspace panel begins at **y = 700px on a 720px-tall screen**. Expanded it begins at y = 1122, and **59% of the page is header**. Nine persistent bands stack before any game content:

Band	Height	Cumulative
Masthead	48 px	70 px
Player identity strip	18 px	98 px
Command-center status bar	45 px	163 px

Band	Height	Cumulative
Eight-metric summary grid	71 px	264 px
Workspace group buttons	39 px	332 px
Workspace tab row	53 px	392 px
Budget strip (3 lines)	107 px	519 px
Simultaneous plan bar	66 px	627 px
Review this month (collapsed)	—	700 px

Clicking a destination scrolls you to something you cannot see. Collapse the masthead and identity strip into one 40px bar once a campaign is running; cut the summary to three headline metrics; move the budget strip into Operations where it is used. Target under 200px.

HIGH

16. Design tokens are bypassed five to one

Measure	Count
var(--token) uses	142
Hardcoded hex literals	730
Distinct hex colors	469

usability.css — the newest work — has 42 hex literals and 2 token uses, and introduces a modern blue SaaS palette on top of base.css's retro monospace paper aesthetic. **Two design languages, neither winning.** Retheming, dark mode or a contrast pass is currently impossible; collapse the palette into tokens *before* any visual work.

MEDIUM

17. The interface explains itself by denial

304 sentences in player-facing strings are built around *is not / are not / does not / never / cannot*. A single month's result reads:

"Collections recovered \$0 principal (not income). Scheduled principal returned \$410,336 to cash (not profit). Former-customer term maturities paid out \$0 (not an expense). Production profit excludes separately reported head-office costs."

The diagnostic

8 of your 34 help topics are misconception corrections: "Research is not the same as implementation", "Applications are not yet activated accounts", "Market influence is not deposit ownership", "Insurance agency is not underwriting"... When a quarter of your help exists to explain that things are not what they are labelled, **the labels are wrong**. Each rename deletes a help topic and a pile of disclaimers. There is also **no help topic explaining Enterprise Value** — the 22-term polynomial that decides who wins.

MEDIUM

18. “Optional systems” is a ladder pretending to be a menu

15 checkboxes that look like a menu are one linear chain of 24 systems. Clicking **Financial Group preview** produces a dialog listing **22 forced dependencies** — including, verbatim, “*Advertising attribution: on — required dependency*”. A bank’s parent-company capital structure does not depend on ad attribution; the chain is build order, not causality. Replace it with 3–4 independent bundles presented as difficulty tiers.

Part 9 · Banking depth and realism

The bank models one risk, sells nine fixed products, and cannot develop anything new. These are the gaps that make it read as a spreadsheet rather than an institution.

HIGH

19. Commercial lending is disconnected from commercial banking

This is the sharpest structural problem in the simulation. Business relationships and commercial loans are produced by different departments and never meet:

```
bizGain = (a.business + commercialFacilities*0.4) * ... -> relationship COUNTS loanGain = a.lending * 185000 * loanMult * training <- lending staff ONLY commercialIncome = business*760 + merchant*650 -> fee income ONLY
```

- **A business banker cannot originate a business loan.** They generate relationship counts that pay a flat monthly fee and nothing else.
- **“Middle-Market Lending” is not commercial banking.** It is one of three allocation percentages inside a retail origination pipeline staffed by a.lending. Winning a commercial relationship does not produce a middle-market loan, and originating a middle-market loan does not require a relationship.
- **The Commercial Banking Office has the highest loan capacity of any building** (430,000 vs retail’s 250,000) — but nothing about the commercial *franchise* feeds origination. The building raises a ceiling the commercial book never fills.
- **Business relationships never churn.** Only a Commercial Client Raid removes one, and that fires in 2–4% of turns. They accumulate permanently, which is why the all-in-business allocation returns \$235,493/month — roughly 8× the net income of the entire loan book, with no funding, capital or credit risk attached.

A commercial bank whose commercial bankers cannot lend is a fee brokerage with a bank attached. Connecting bizGain to a separate businessLoans book — short, floating-rate, revolving, lumpy — is the change that makes the institution coherent, and it is the honest way past the origination ceiling in Part 3.

HIGH

20. Loan “types” are three multipliers on one pipeline

There are three credit families, and they do differ where it matters — term structure (120 / 48 / 24 months) drives a 4.4× swing in the sustainable book. But they are not products:

	What the game has	What is missing
Origination	One shared pipeline split by a % allocation	Separate channels, separate officers, separate demand

	What the game has	What is missing
Underwriting	A global 3-position multiplier	Individual credits, collateral, LTV, debt-service coverage, an approve/decline decision
Loan families	Mortgage · Middle-market · Consumer	HELOC, auto, credit card, SBA, construction, agricultural — construction and ag being the two that most define real community-bank risk
Losses	loans × 0.25%/mo, smooth and immediate	Lumpy, lagging, concentrated losses; a provision/allowance account

`collections.js` states outright that there is no “*full allowance/reserve model*”, so charge-offs strike equity directly with no provisioning. The delinquency ageing in `creditPerformance` is genuinely good and is the right foundation to build on.

HIGH

21. There is almost no research or product development

The entire sellable catalogue is **nine fixed SKUs**, and only two of them can ever be created:

Line	Products	Lifecycle	Player control
Retail	Essential · Rewards Checking · High-Yield Savings	none → build partner → retired	Deploy or license; retire; price two of the three
Credit	Mortgage · Middle-market · Consumer	always present	A percentage allocation
Business	Relationship · Treasury · Entrepreneur	always present	Pick one of three

- **There is no upgrade path.** A product is off, on, or retired. You cannot improve one, add features, create a variant, or tier it. `retired: true` is the only state change available after launch.
- **“Research” is five spend meters.** The strategy tree (Part 5) has no prerequisites across branches and no exclusive routes — `strategyLevel` is cumulative spend crossing thresholds.
- **The pricing control is on the wrong products.** `pricingBp` is initialised as `{essential: 0, rewards: 0}` — **High-Yield Savings has no rate control**, despite being the one product whose entire identity is its rate (`sensitive: .16`, described as “expensive and highly rate-sensitive funding”). You can price the free checking account but not the high-yield savings account.

Cheapest meaningful fix: give each deployed product 2–3 investable tiers (Rewards Checking → Premium Rewards → Private Client). Same three SKUs, real progression, and it gives the pricing system something to act on. Start by adding the missing High-Yield rate knob — that is a missing control, not a new system.

HIGH

22. The bank carries one risk category out of four

Audited across engine, content and UI for operational-risk concepts:

Concept	Hits	Concept	Hits
overdraft / NSF	0	AML / BSA / KYC	0
chargeback / dispute	0	FDIC / deposit insurance	0
ACH / wire transfer	0	loan-to-deposit / NIM	0

Concept	Hits	Concept	Hits
interest-rate risk	0	teller / branch hours	0
fraud	4 — all one event card	CRA	0

The only fraud in the game is a single monthly event, SUSPICIOUS TRANSACTION PANIC → *Freeze and investigate / Keep processing*. **Credit charge-offs are the only loss the bank can take.** Real banks carry credit, interest-rate, liquidity and operational risk; this models one.

Two consequences worth naming:

- **No interest-rate risk.** `macro.rate` moves 3.0–6.25 and changes income and cost immediately and symmetrically. There is no duration, no repricing gap, no asset/liability mismatch — so funding 120-month fixed mortgages with overnight deposits, the single most dangerous position in banking and the one that actually caused failures in 2023, costs nothing.
- **Missing fee income.** Overdraft and NSF are typically **15–25% of a community bank's non-interest income** and are entirely absent, along with interchange (one passing mention), wire fees and safe-deposit.

Why operational risk is the one to add first

It fixes a problem you already have. Operations bankers are the only staff who generate no revenue (Finding 6) and their charge-off reduction is worth ~\$2,250/month against a \$26,000 opportunity cost. Give them fraud, EFT disputes and error losses scaled to transaction volume and merchant count, and the department earns its payroll. **It also gives merchant services a downside** — chargebacks scale with merchant count, so the counter that currently only goes up finally has a cost.

MEDIUM

23. Merchant services is a counter with no screen

Merchant grows from business staff, commercial facilities and technology, and pays `merchant × $650/month`. That is the whole system.

- **There is no merchant panel anywhere.** It appears as a number in the summary strip and the final scorecard — you cannot see the book, its drivers, or what it is worth.
- **No merchant decisions exist** — no processing rates, terminals, settlement, interchange or chargebacks. The only levers are Business staffing and the Treasury product option ($\times 1.16$).
- **No attrition**, as with business relationships.

MEDIUM

24. The origination ceiling is real, but its growth path is priced out of reach

The cap is not arbitrary — it is a staff-and-facilities capacity model, and it *can* grow. The problem is that nobody can afford to grow it:

Retail offices	Loan capacity	Build cost	Upkeep	Staff needed
2 (typical)	\$0.60M/mo	\$1.3M	\$44K/mo	6 FTE
4	\$1.10M/mo	\$2.6M	\$88K/mo	12 FTE
8	\$2.10M/mo	\$5.2M	\$176K/mo	24 FTE

Retail offices	Loan capacity	Build cost	Upkeep	Staff needed
18 (theoretical max)	\$4.60M/mo	\$11.7M	\$396K/mo	54 FTE

Measured across 48 campaigns: **2–4 offices and 6–10 total staff**. The route to a larger book exists on paper and is never travelled. Combined with the fact that the ceiling is never shown on screen (Finding 5), it reads to the player as an arbitrary wall.

It also models the wrong constraint. Real banks of this size are limited by capital and by loan demand, not by how many applications an officer can process. Capital is already modelled well here through the tier system — leaning on that instead of a hard `min()` would be both more realistic and more legible.

Part 10 · Construction and layout

MEDIUM

25. The flagship building is strictly worse than two basic ones

	1 Integrated Financial Center	2 Retail branches	Winner
Build cost	\$1,400,000	\$1,300,000	2 retail
Monthly upkeep	\$48,000	\$44,000	2 retail
Staff quarters	32 (8 FTE)	24 (6 FTE)	2 retail
Deposit capacity	700,000	900,000	2 retail
Loan capacity	600,000	500,000	Financial Center
Service capacity	3.0	3.0	tie
Advisory capacity	2	0	Financial Center

It wins only on loan capacity — the worst asset in the game — and on advisory, which feeds a wealth system gated behind a licensed subsidiary that does not exist. Efficiency across the full catalogue:

Model	Capacity per \$	Capacity per upkeep	Capacity per FTE
Retail	1.08	31.8	233,333
Commercial	0.94	30.0	165,000
Digital	0.90	39.2	188,000
Financial Center	0.93	27.1	162,500
ATM	0.47	15.6	140,000
Regional Hub	0.27	8.1	71,429
Wealth office	0.10	3.2	26,667

Seven models: two are good, one is situational, four are traps. The wealth office is **10× worse than retail** on capital efficiency.

HIGH

26. The capacity preview reads +\$0K for every building

Every project card in the build menu shows:

ON COMPLETION · facility cost +\$26K/month · deposit capacity +\$0K · loan capacity +\$0K

Both the \$644K retail branch and the \$1.61M financial center report **+\$0K**. The expense delta computes correctly, so `regionalProjectPreview` is running — only the capacity half returns zero. Ramp-up does not explain it: a new office starts at $\text{ramp} = (0+1)/4 = 0.25$, so it should read 25% of capacity, not nothing.

The single number a player would choose a building on is zero for every building. This is also why Finding 25 is invisible in play — there is no way to discover that two retail branches beat the flagship.

MEDIUM

27. The same building has two names, and two buildings share one name

Model key	Build menu	Office list
retail	Full-Service Financial Center	Retail branch
financialCenter	Integrated Financial Center	Financial center
commercial	Commercial Banking Office	Commercial banking office
digital	Digital Advisory Studio	Digital advisory studio

You buy a *Full-Service Financial Center* and it appears in your network as a *Retail branch*. And two buildings carry “Financial Center” — the \$644K one and the \$1.61M one — with the cheaper carrying the grander name. Build cards also leak raw internal keys: `FINANCIALCENTER_MODEL` and `REGIONALHUB_MODEL`, from `p.facility.toUpperCase()` in `ui/projects.js`.

MEDIUM

28. Building and managing offices are two nav groups apart

- **You cannot build from the construction screen.** Office management lives in *Customers & markets* → *Markets*; building lives in *Run the bank* → *Operations* → *Projects & hiring*. `ui/facility-network.js` has zero references to `PROJECTS`.
- **Defending a market you are losing spans three tabs in two groups:** see the market (*Customers & markets*), read the exit countdown (*Growth & competition* → *Intelligence*), choose a defense (*Growth & competition* → *Competition*).
- **“Products” is both a tab and a desk** — a top-level Products workspace and a “Products & funding” desk inside Operations.
- **Credit and lending are separated:** loan quality and collections sit in Bank & finance; the allocation that creates those loans sits in Operations.
- **The monthly plan — the thing you do every turn — is two levels down** inside Operations.
- **Two office panels on one screen:** `#facilityNetworkPanel` (“OFFICE NETWORK”) and `#facilityLifecyclePanel` (“OFFICE CONDITION & STAFFING · 1 operating sites”).

MEDIUM

29. Commit ceremony is the biggest quality-of-life drag

Verb	Buttons in the UI
Stage	30
Preview	13
Prepare	10
Confirm	6
Undo	6
Adopt	2

Four different verbs for committing. Managing a single office needs six buttons: *Stage office conversion* · *Preview renovation at this office* · *Preview settings* · *Prepare staffing proposal* · *Stage settings and renovation order* · *Discard unsubmitted lifecycle changes*.

And previews go stale constantly — nine distinct “do it again” messages exist, because every desk shares one monthly draft, so touching anything invalidates previews on unrelated desks:

“The form changed. Preview it again.” “The proposal is stale. Prepare the current plan again.”
 “Shared plan changed. Your unstaged values are retained; preview again before adoption.”

Fix: collapse Preview/Prepare into one live-updating panel that recomputes instead of going stale; use one verb for committing; auto-restage anything still valid after an unrelated change. For a game played in office breaks, this is the friction that costs the most.

LOW

30. Engine plumbing on screen

- **Internal office IDs are shown to the player:** the office table reads *Downtown* / `bank-3c0d66e4:office:1`. A hash fragment means nothing to a human — “Downtown — Retail branch #1” would.
- **“\$177,188 / \$300,000”** under the header *Deposit* / *loan capacity* reads as a fraction rather than two independent values.
- **Mixed money scales in one list:** \$644K, \$1.61M, \$173K, \$2.13M.
- **Plural bugs continue into construction:** “1 operating sites”, “1 base work units”.
- **“PAID RE-ENTRY AVAILABLE”** is a hardcoded static banner (`#unlockText`) shown whether or not you have lost anything. It never names a market, a price or a countdown.

MEDIUM

31. Explanations that defend numbers instead of explaining them

From `ui/relationship-offers.js` — 522 characters:

“The share reserves physical Retail time after retention. Additional function allocations and purchased work can add capacity. Remaining sales capacity already accounts for onboarding and all other reservations; it is not deducted again. Vendor fees are already included once in the shared plan, separately from conversion expense.”

Two of four sentences exist purely to say something is *not* double-counted. The player never suspected double-counting — **the text creates the anxiety it then resolves**. There are eight of these reassurances across

the UI.

- `ui/recovery.js` carries a seven-item exclusion list: *“Excludes uncertain executive effects, rival actions, opportunity awards, regulatory sales, future project benefits and changes in demand.”*
- `ui/services.js` uses three profit definitions in one sentence: *“Desk net is before shared payroll; bank profit includes payroll and the immediate effect of taking bankers out of sales, but not future lost relationships.”*
- The workforce teaching blurb spends five sentences and four caveats to say “training takes a banker off the floor.”

The pattern is always the same: a number is shown, then defended. Show fewer numbers and defend none of them — move the reconciliation behind a “how is this calculated” disclosure for the people who want it.

Part 11 · Gameplay dynamics

This part is measured, not read from formulas. Each row below holds one decision constant for an entire campaign and compares the final score against the AI baseline — 12 campaigns of 72 months per row, identical seeds throughout.

Read findings 43–46 at the end of this part alongside the table below. They measure how much of a campaign result is skill and how much is chance, and they change how every figure here should be interpreted.

CRITICAL 32. The decision space is three decisions wide

Decision held constant for the whole campaign	Score	Delta	% of baseline
Never pursue opportunities	1202	−496	−29.2%
Never bid on service contracts	1239	−459	−27.0%
Morale-safe staff allocation	2146	+448	+26.4%
All-in lending allocation	1411	−287	−16.9%
No research spend	1446	−252	−14.9%
Lending → growth	1499	−199	−11.7%
Deposits → aggressive	1535	−163	−9.6%
Never start projects	1573	−125	−7.4%
Lending → conservative	1811	+113	+6.6%
Always executive option B	1800	+102	+6.0%
Business → entrepreneur	1619	−80	−4.7%
Deposits → margin	1772	+74	+4.3%
Credit → consumer	1634	−64	−3.8%
Capital → reinvest	1747	+48	+2.9%
Never attack or defend	1722	+24	+1.4%
Business → treasury	1709	+11	+0.7%
Retail product → rewards	1698	+0	0.0%

- **The two biggest levers are the two smallest panels.** Opportunities (–29%) and contract bids (–27%) decide the campaign. Opportunities is a strip inside the Markets workspace; contract bidding is buried further still. Neither has a workspace of its own.
- **The Competitive War Room is worth +1.4% — noise.** Eight actions, concealed simultaneous resolution, its own workspace tab, and the game's *name* — and switching it off entirely leaves you very slightly better off.
- **Product choice barely registers.** The “PRODUCT & BALANCE-SHEET DESIGN” panel is introduced as *“Choose what the bank actually sells... these products alter customer growth, funding cost, fee income, loan production, and credit losses every cycle”* and moves the outcome by 0–4%. Retail product choice is **exactly zero**.

CRITICAL 33. The dominant strategy is disengagement

Read the positive column: conservative lending, margin deposits, always take the free executive option, never attack. Read the negative: growth lending, aggressive deposits, all-in on a department, paying for things.

Always clicking executive option B beats always clicking A by 12.6 points of score. Option A is nearly always the “spend cash” branch. The one mandatory decision every month is therefore best answered by never spending — so a player who reads none of the flavour text and always clicks the right-hand button outperforms one who engages with it.

Every paid branch of every decision is net-negative. That is the through-line of Part 3: with deposits out-earning loans risk-free, the correct play is to accumulate and decline. The gameplay measurements are simply that economics showing through.

CRITICAL 34. Six fixed settings beat the hardest AI 85% of the time

A scripted player was given five settings applied blindly on turn 1 and never revisited — morale-safe allocation, conservative lending, margin deposits, always option B, never attack — with everything else left to the bot:

	Result
Campaigns against Chairman AI	20
Human wins	17 / 20 (85%)
Median margin	+396
Losses	3

No market reading, no situational judgment, no adaptation. The hardest opponent in the game loses five times out of six to a fixed configuration.

CRITICAL 35. The AI cannot play its own game

Its default opening allocation is {service: 2, business: 2, lending: 3, operations: 1}. Morale requires $\text{service} \geq 5$ and $\text{operations} \geq 2$:

$\text{change} = 2 - \max(0, 5-2) - \max(0, 2-1) = -2$ per month, from turn 1

The AI's own default guarantees morale decay forever, which is why median morale ends at 12 out of 100 (Finding 40). And the morale-safe allocation is the single largest positive lever in the entire sensitivity table at **+26.4%** — the AI never takes it, at any difficulty.

This is close to a one-line fix with the largest effect-per-character in the whole review. It converts the hardest opponent from “loses to five fixed settings” into something that requires attention, and it lifts the morale system out of permanent collapse at the same time.

HIGH

36. The campaign hardens at month 24 and is over by 48

Month	Leader here also wins	Mean lead	Lead changes remaining
6	60%	77	1.5
12	65%	116	1.1
18	70%	221	0.7
24	75%	334	0.7
36	80%	388	0.4
48	90%	431	0.3
60	90%	444	0.1

Final margin: **median 407** on a ~1,700 baseline — a 24% blowout. **Zero of twenty campaigns finished within 5%.**

So the shape is: genuinely open for about 20 months, hardening through 40, and from month 48 a 90%-determined outcome with essentially no lead changes left — and in pilot mode it then **never ends** (Finding 2). You keep playing a decided game indefinitely. For a two-player office game this is the most consequential dynamic in the review: the last third is a formality and the format has no way to stop.

Scenario choice changes how decisive a campaign is, not how it is played — mean final margin runs Balanced 232, Rate Shock 318, Growth Frenzy 492, Regulatory Siege 683. The Regulatory scenario produces three times the blowout of Balanced.

CRITICAL

43. The variance is larger than the skill signal

Two **identical** players running the **identical** strategy were run across 32 campaigns. No skill difference exists between them, so any final margin is pure chance.

Matchup	Median margin	Mean	Max	SD
Identical players, identical strategy	658	676	1805	762
Six fixed rules vs Chairman AI (a large skill edge)	396	—	—	—
AI vs AI in the lead study	407	—	—	—

Playing well moves you about 400 points. Playing identically moves you about 658. Seat 0 won 47% of campaigns, so this is not a seat advantage — it is variance.

Why this matters more than it sounds

A player who loses 1,400 to 2,058 will read that as being thoroughly outplayed. At equal skill that exact result is a coin flip. **The final margin looks decisive and carries almost no information about who played better** — which, for a game played against a colleague, is the single thing most likely to make it feel unfair.

A large skill gap still wins consistently — the six-rules player took 85% of its campaigns — because both seats share one game world and much of the noise is common to both. But between players of *similar* ability, the scoreboard is close to a lottery.

CRITICAL 44. Opportunities are both the biggest lever and the biggest noise source

Same identical-player setup, disabling one subsystem at a time for both seats:

Configuration	Median margin	SD
Everything on (baseline for this run)	495	552
Opportunities disabled for both	248	364
Contract bids disabled	323	426
Competitive actions disabled	381	538

Turning opportunities off halves the variance. Absolute numbers differ from Finding 43 because this harness consumes the random stream in a different order; the valid comparison is within this table.

So the decision worth −29% of your score if you skip it (Finding 32) is also the mechanic generating half the randomness. **The game's most important decision is also its most random.** Three opportunities spawn each month in markets chosen by a uniform `pick(open)`, with a ± 5 roll on top of your power score.

In fairness to the design, the *contested* resolution is well tuned — simulated over 200,000 draws:

Power edge in a contested draw	Wins
5 points	~100%
2 points	~82%

That part rewards investment properly. The randomness lives in *which* opportunities appear and *where* — not in who wins a head-to-head.

HIGH 45. The opportunity formula has three non-functional terms

`opportunityPower` sums roughly nine terms. Three do not work:

```
... + p.stats.reputation/22 + p.stats.morale/35 + p.upgrades.training + p.upgrades.analytics + ...
```

- `p.upgrades.training` and `p.upgrades.analytics` are permanently 0, because those projects are retired (Finding 38). Two dead terms sit inside the game's single most important formula.
- `reputation/22` is a constant. Reputation saturates at a median of 100 (Finding 42), so it contributes a flat +4.55 to both players, always — it cannot differentiate them.

- morale/35 **is crippled**. It should contribute about 1.9 at the opening morale of 68; median morale is 12, so it contributes 0.34. The morale collapse quietly costs about 1.6 power every month.

The formula reads as a rich nine-factor calculation and functions as roughly five.

MEDIUM

46. Dead zones in opportunity odds, and a hidden contract gate

Uncontested opportunities clamp at both ends: `chance = clamp(.32 + power/38, .42, .90)`

Power	Win chance	
0 – 3.8	42%	investment below this does nothing
10	58%	
18	79%	
22.04 and above	90%	investment above this does nothing

The useful band is **3.8 to 22.0**. Since reputation/22 alone contributes 4.55, the floor is cleared for free — but a developed bank sits at 18–20, so it spends much of the late game near the cap where further investment is wasted.

Contract bidding has an undisclosed eligibility gate. `if (p.allocation.business < 1 || ...) continue` — with zero business bankers you cannot win *any* service agreement, incumbent or challenger. Since contract bids are worth 27% of score (Finding 32), a player who moves business to zero silently forfeits roughly a quarter of the game, with no warning anywhere in the interface.

Part 12 · Progression and dead systems

Five progression systems were built, superseded, and left wired into the economy. They all trace to the same root, and they are why several of the numbers above look stranger than they should.

HIGH

37. Four of eight mandates are unachievable; three are automatic

The confidential mandate is worth 100–120 points of a roughly 1,700 final score. Across 48 player-campaigns at 90 months:

Mandate	Assigned	Achieved	Rate
Commercial Powerhouse	5	5	100%
Deposit Titan	6	6	100%
Clean Books	10	9	90%
Market Dominance	11	2	18%
Digital First	5	0	0%
Earnings Machine	6	0	0%
Physical Empire	3	0	0%
Employer of Choice	2	0	0%

Because the mandate is randomly assigned and secret, **which one you draw largely decides whether you collect the bonus**. That is a coin flip on the scoreboard, not a strategy.

HIGH

38. Retired upgrades leave dead terms throughout the economy

All five upgrade projects are marked `legacy: true` with the description “*Legacy save compatibility*”, and `institution.js:161` sets `retired: !!def.legacy`. They cannot be bought in a current campaign — measured across 24 player-campaigns, training and technology were **min 0, median 0, max 0**.

Two mandates still require them — **Employer of Choice** (“training level 2+”) and **Digital First** (“technology level 2+”). That is why both sit at 0%.

Worse, **18 references to permanently-zero variables remain in the two core economy files**:

```
training = (1 + u.training*.08) * ... -> always 1.0 tech = 1 + u.technology*.08 + u.analytics*.035 +
... -> 2 of 3 terms dead revenue = ... + u.technology*13000 + u.wealth*18000 -> both dead efficiency
= (1 - u.operations*.06) * ... -> always 1.0 baseScore = ... +
(technology+training+analytics+wealth+operations)*13 -> always 0
```

An entire term of the win condition is permanently zero, and the operating formulas look substantially richer than they are.

HIGH

39. Cumulative earnings are negative — every investment is expensed

Measure (90 months)	Min	Median	Best case
Cumulative earnings	−\$2.4M	−\$670,119	+\$62,508

The Earnings Machine mandate needs **+\$2.2M**; the best observed result is 35× short. The cause is in `accounting-adapter.js:26`:

```
if(n<0){ provideCash(p,-n); post(p.accounting, source, {cash:n, equity:n}, n) }
```

Any cash outflow reduces equity one-for-one. There is no fixed-asset account — a \$650,000 branch is a \$650,000 immediate loss and the building carries zero balance-sheet value forever after. Real banks capitalise premises and depreciate over decades. As written, **growth always reads as failure on the earnings line**, which quietly reinforces the vault strategy in Part 3.

HIGH

40. Morale is a losing battle by construction

```
change = base(2) − max(0, 5 − a.service) − max(0, 2 − a.operations)
```

Holding morale steady requires **seven of your eight opening bankers** (service 5, operations 2), leaving one for business and lending combined — and operations bankers generate no revenue at all (Finding 6), so two of those seven are pure cost.

Measure at 90 months	Min	Median	Max
Staff (opens at 8)	2	6	14
Morale (opens at 68)	0	12 / 100	100

Staff *falls* from 8 to a median of 6, putting the seven-staff break-even permanently out of reach. Morale feeds bizGain at $(1.25 + \text{morale}/120)$, so the drop from 68 to 12 costs about 26% of commercial acquisition. It is a silent decay nobody can fight — and the one lever that would offset it, the Staff Development Academy, is retired.

MEDIUM

41. Three of five doctrines never occur

Doctrine is never chosen. `player()` defaults it to 'community', then `syncDoctrine` recomputes it each turn from allocation shares, capability spend and facility mix. Across 48 player-campaigns:

Doctrine	Times it occurred	Why
Community	38	Free head start: <code>facShare('retail')</code> is positive from turn 1
Digital	10	Reachable through digital capability spend
Commercial	0	Needs business allocation above 25% AND commercial spend; measured avg only \$244K
Efficiency	0	Requires over-allocating to Operations — the department that earns nothing
People	0	Depends partly on <code>capShare('acquisition')</code> , measured at \$0 in every game

Five doctrines with distinct bonuses, and the player cannot pick one, is never told how they are derived, and can realistically reach two.

MEDIUM

42. Four stats saturate and stop mattering

Stat	Min	Median	Max
Reputation	78	100	100
Momentum	82	100	100
Influence	61	98	100
Compliance (risk)	0	0	24

Reputation, momentum and influence pin at their ceilings; risk floors at zero. Four stats that feed the score and several decisions become constants after the early game — and it is why **Clean Books is achieved 90% of the time**: the risk it measures is almost always already 0.

Part 13 · Implementation record

Work has since landed on `feat/v3-usability`. This part records what changed, what it measured at, and what could not be done there.

The constraint that governs everything

This branch cannot change the simulation

tests/usability_engine_boundary.test.js pins the assembled engine to SHA-256 cc229f83... with the note: *“Immutable reviewed V3.1 c721ede engine. This UI release has no authority to retune simulation. Do not update the digest to accommodate accidental drift.”* Any edit under `src/engine/` or `src/content/` fails the gate. **Every balance change in Parts 3, 4, 5 and 11 therefore belongs on a separate branch**, taken after this one merges.

This is a real boundary doing real work, not an obstacle. It caught a capacity fix that had been written into an engine module; moving it to the UI layer produced better-scoped code that behaves identically.

Fixed and verified

Finding	Was	Now
12 · LAN auto-detection never worked	PowerShell parse error, address typed by hand	Detects the LAN address; correct <code>file:///...#lanip=</code> URL
26 · Build preview read +\$0K for every model	Every building looked identical and worthless	Rated capacity per model, verified against catalogue × market profile
15 · 700px of chrome before any content	No workspace visible on a 720px screen	247px — nine bands became three
14 · The comeback arc was invisible	11% share rendered identically to 45%	Exit countdown and re-entry price on the map tiles
46 · Contract eligibility gate undisclosed	Zero Business staffing silently forfeited every agreement	Warned in Review this month, incumbents included
32 · Biggest levers had least presence	Opportunity pipeline sat below the map	Promoted to the top of the Markets workspace
13, 30 · Copy and formatting defects	9 plural bugs, “1 operating sites”, “\$0K”, raw office ids	All fixed; capacity column split; ids humanised
17 · Explanations by denial	304 negation-style sentences	159 — three worst blocks rewritten
17 · No help topic explained how you win	34 topics, none covering Enterprise Value	Topic added, targeted at the score itself

The chrome wall, measured

Viewport	Before	After
1280×720, review collapsed	700 px	247 px
1280×720, review expanded	954 px	500 px
900×720	~830 px	454 px

Viewport	Before	After
700×720	—	546 px

No horizontal overflow at any width. The masthead hides once a campaign is running; seat identity, status and three headline figures share one command bar; the shared spending strip moved into Operations; the other five figures sit behind an inline disclosure with every metric id preserved.

A defect found while building the countdown

The exit threshold is not fixed. Under regional operations an office closes after **six** cycles below 12% share (`regional.js:78`); under base and accounting-pilot rules it closes after **three**. The existing threat board hardcodes `{streak}/3` in `session.js`, so **regional players are told they have one cycle left when they have four**. The new map tiles use the correct per-ruleset denominator; the threat board itself is an engine file and remains wrong.

Blocked by the engine boundary

Item	Blocked by
Threat-board severity 'watch' → 'hot'	<code>src/engine/session.js</code>
Threat-board /3 denominator bug above	<code>src/engine/session.js</code>
Re-entry protection window on tiles	<code>reentryUntil</code> is not in the public view
Rename <i>Market Leader</i> → <i>First Market</i> (Finding 13)	<code>src/content/catalog.js</code> + <code>behavior-golden.json</code>

The milestone rename is worth a note: it is display-only — two game states are byte-identical once the string is normalised — but the golden fixture hashes whole game states, so the name appears in seven fixtures. It needs a sanctioned regeneration recorded in the changelog, on a branch that is not engine-frozen.

Verification

Check	Result
<code>tools/check.js</code> fast gate	152 / 152 groups, 0 failures
Engine hash	cc229f83... unchanged
Portable build	Current, 143 source files
Full release gate	Run in progress at time of writing

Three defects were introduced and caught during the work itself, all by measurement rather than inspection: a `min-height` floor that made the merged bar taller than the three it replaced, grid columns whose `minmax()` floor exceeded their flex box and rendered outside it, and a string literal truncated mid-markup by a plural fix. The last was caught by the build; the first two only by measuring the rendered layout.

Nothing in this part changes the analysis in Parts 1–12. The economic findings — deposits out-earning loans, the dominant policy options, the flat loan ceiling, the variance exceeding the skill signal — are all untouched, because none of them could be touched here.

Part 14 · Prioritised roadmap

Phase 0 — four small fixes with outsized effect (hours) · 2 of 4 landed

Change	Where	Effect
Give the AI a morale-safe default allocation	<code>engine/ai.js</code>	Largest effect-per-character in the review: +26.4% of score, and lifts morale out of permanent collapse.
Fix the capacity delta in the build preview	<code>regional.js:48</code>	Build cards stop reading +\$0K; construction becomes choosable.
Remove the carets from the PowerShell command	<code>OPEN_BRANCH_WARS.bat</code>	Two-computer play works out of the box.
Rename <i>Market Leader</i> to <i>First Market</i>	<code>catalog.js:156</code>	Stops both players 'winning' the same exclusive rank.

Phase 1 — fix the economy (days)

Everything else multiplies whatever economy it sits on. Do this first.

Change	Where	Effect
Cut <code>depositIncome</code> below the loan spread, or tie it to deployment	<code>operations.js:14</code>	Ends the vault strategy at its root.
Win Deposits raises deposit capacity, not just cost	<code>operations.js:13</code>	Removes a strictly-negative option.
Growth lending raises <code>loanCapacity</code>	<code>operations.js:13</code>	Makes lending standards a real trade.
Drop the cash coefficient below the loans coefficient	<code>territories.js:7</code>	Scoreboard stops arguing with the theme.
Include <code>a.operations</code> in per-banker revenue	<code>operations.js:14</code>	Operations stops being a pure tax.

Phase 2 — restore the comeback (days)

- Allow branch projects into exited markets at a share-scaled premium; clear `exited[index]` on completion.
- Ungate the re-entry grace path from `p.accounting`.
- Taper or cap `franchiseDividends`.
- Add a scored ending at a fixed horizon in pilot mode — now load-bearing, since a fully recoverable game has no other terminator.
- Surface the exit countdown: 'hot' severity, on the map, with the re-entry price on withdrawn tiles.
- Remove the carets from `OPEN_BRANCH_WARS.bat`; rename *Market Leader* → *First Market*.

Phase 3 — build the commercial bank (1–2 weeks)

- Add `businessDeposits` / `businessLoans` driven by `bizGain`, tracked separately from retail.

- Commercial deposits excluded from `rateSensitiveDeposits`, with a concentration penalty when too few relationships hold too much.
- Commercial loans on short floating terms that revolve rather than amortize — the honest way past the \$10M–\$25M ceiling.
- Monthly relationship attrition, so business bankers must retain rather than accumulate.
- Reframe the three credit products around term structure, not “volume”.

Phase 4 — make it legible (1–2 weeks)

- Collapse the chrome wall to under 200px; move the budget strip into Operations.
- Consolidate 469 hex colors into the token set; pick **one** visual language.
- Show the loan-book ceiling; add an Enterprise Value help topic and score breakdown.
- Give free event branches an expected-range preview; add a capital-ratio delta to the lending control.
- One money formatter, one ratio formatter, correct plurals throughout.

Phase 4b — fix construction (days) · preview and naming landed

- Fix the `regionalProjectPreview` capacity delta so build cards stop reading +\$0K, and add a side-by-side comparison of all available models.
- One name per building: reconcile `PROJECTS` names with `FacilityLifecycle.CATALOG` names, and stop printing raw model keys.
- Rebalance the Financial Center so it beats two retail branches on something a player wants.
- Put building and managing on the same screen.
- Replace internal office IDs with human names; split the “deposit / loan capacity” column in two.

Phase 5 — add the missing risk (2–3 weeks)

- **Operational risk first** — fraud, EFT disputes and error losses scaled to transaction volume and merchant count. This gives the Operations department a reason to exist and gives merchant services a downside.
- **Interest-rate risk second** — connect the existing 24/48/120-month loan terms to the existing rate-sensitive deposit balances, so funding long fixed assets with hot money hurts when rates move.
- Add overdraft/NSF and interchange fee income; add a provision/allowance account so losses are anticipated rather than instant.
- Give High-Yield Savings a rate control, then add 2–3 investable tiers per deployed product.
- Skip AML/BSA/CRA for now — the existing `compliance` stat already carries that abstraction.

Phase 5b — fix the shape of a campaign (1 week)

- **Promote opportunities and contract bids to first-class.** They already decide the game (–29% and –27%); give them the space the war room currently occupies.
- **Make the war room matter or cut it.** +1.4% for the game's headline feature is untenable — scale action costs to bank size, or fold the drama into contested opportunities and delete the rest.

- **Cap the campaign near where it is decided** — 36 to 48 months rather than open-ended. This is the same change as the Phase 2 scored ending, sized from the measured data.
- **Add pressure between months 24 and 48**, where the lead currently calcifies: catch-up mechanics, escalating events, or the Part 2 comeback system doing real work.
- **Retire the retired upgrades properly** — either restore them as purchasable or strip their variables from the economy and rewrite the two mandates that depend on them.
- **Capitalise premises** so building converts cash to an asset instead of destroying equity.

Phase 5c — close the gap between skill and noise (1 week)

If the variance stays larger than the skill signal, better play still will not reliably win — and that is what decides whether two colleagues keep playing.

- **Make opportunity placement partly deterministic.** Keep three per month, but let the player influence which markets they appear in — through focus, branch presence, or a pipeline-development investment. Today `pick(open)` is uniform random. This converts the largest noise source into a strategic one without removing the drama.
- **Report the result honestly.** Show the final margin with a confidence band, or score the campaign on something less noisy than raw enterprise value. A 658-point loss should not read the same as a 658-point outplaying.
- **Clean up** `opportunityPower` — drop the two dead upgrade terms and replace the saturated reputation term with something that still varies late in a campaign.
- **Surface the contract eligibility gate** so moving business staff to zero does not silently forfeit a quarter of the score.

Phase 6 — deepen the tree and the competition (2–4 weeks)

- Convert strategy tiers from cumulative spend to **exclusive choices**; add cross-branch prerequisites.
- Repoint the Acquisition branch at the re-entry system so it becomes the comeback branch.
- Scale competitive-action costs to bank size; widen trigger conditions.
- Resolve deposit contests **per market**; remove the `competitionScale` penalty on larger maps.
- Replace the 24-system ladder with 3–4 independent bundles.

Closing assessment

The simulation underneath is better than the game on top of it, and the design intent is sound — the comeback arc, the term structure, the capital-tier supervision and the 51-cycle decline are all genuinely good banking and genuinely good drama.

One caveat on all of the above: the measured variance between two identical players (Finding 43) is larger than the gap a large skill edge produces. Balance changes will improve the quality of decisions, but until the noise is reduced the scoreboard will keep overstating how decisively one side won.

What holds it back is a small number of coefficients pointing the wrong way. Deposits out-earning loans risk-free is the single line that makes the optimal bank a vault; from there the dominant policy options, the flat loan ceiling and the idle war room all follow. The comeback mechanic you designed is complete — it is blocked by one validator string and one runaway dividend.

Phase 1 and Phase 2 together are roughly a dozen small edits, all with file and line references above. They are worth doing before any new systems, any tech-tree expansion, and the UI overhaul — because each of those multiplies the economy underneath, and changes what the interface needs to show.